

LAB # 4 : Mathematical Expectation

Introduction

In probability theory, the expected value or mean of a random variable is intuitively the long-run average value of repetitions of the experiment it represents. For example, if two coins are tossed 16 times and X is the number of heads that occur per toss, then the values of X are 0, 1, and 2. Suppose that the experiment yields no heads, one head, and two heads a total of 4, 7, and 5 times, respectively. The average number of heads per toss of the two coins is then

$$\frac{(0)(4) + (1)(7) + (2)(5)}{16} = 1.06.$$

Hence, an average is not necessarily a possible outcome for the experiment. For instance, a salesman's average monthly income is not likely to be equal to any of his monthly paychecks.

Let us now restructure our computation for the average number of heads so as to have the following equivalent form:

$$0 \cdot \frac{4}{16} + 1 \cdot \frac{7}{16} + 2 \cdot \frac{5}{16} = 1.06.$$

The numbers $4/16$, $7/16$, and $5/16$ are the fractions of the total tosses resulting in 0, 1, and 2 heads, respectively. These fractions are also the relative frequencies for the different values of X in our experiment.

The expected value is also known as the mean, or first moment.

If X is discrete random variable with pmf f , then for any real-valued function g

$$E[g(X)] = \sum_x g(x)f(x).$$

Similarly, if X is continuous with pdf f , then

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx.$$

The variance of X is

$$\sigma_X^2 = E[(X - \mu_X)^2] = E(X^2) - \mu_X^2.$$

You may find it convenient to print the pdf version of this lab rather than the web page itself. This lab will take one session.

Expectations in R

Example 1: Find $E(X)$ and $E(x^2)$ if X is the outcome of a toss of a fair die.

```
x <- c(1,2,3,4,5,6)
p <- c(1/6,1/6,1/6,1/6,1/6,1/6)
ex<- sum(x*p)
```

```

3.5
# OR
weighted.mean(x*p)
3.5
ex2 <- sum(x^2*p)
15.16667

```

Example 2: Let X be the random variable that denotes the life in hours of a certain electronic device. The probability density function is

$$f(x) = \begin{cases} \frac{20000}{x^3}, & x > 100 \\ 0, & \text{otherwise.} \end{cases}$$

Find the expected life of this type of device.

```

f=function(x) { 20000/x^3 }
pdf <- integrate(f,100,Inf)
print(pdf)
pdf 1 with absolute error < 4.7e-07
print(pdf$value)
[1] 1
ex <- integrate(function(x) x*f(x),100,Inf)
ex 200 with absolute error < 0.0062
ex$value
[1] 200

```

Example 3:

```

x <- c(0,1,2,3,4)
probx <- c(0.41,0.37,0.16,0.05,0.01)
avg <- sum(x*probx)
avg
varix= sum(x^2*probx)-(weighted.mean(x,probx))^2
varix
# output
[1] 0.88
[1] 0.8456

```

Moments in R

The r th moment about the origin of the random variable X is given by

$$\mu'_r = E(X^r) = \begin{cases} \sum x^r f(x), & \text{discrete} \\ \int_{-\infty}^{\infty} x^r f(x) dx, & \text{continuous.} \end{cases}$$

Since the first and second moments about the origin are given by $\mu'_1 = E(X)$ and $\mu'_2 = E(X^2)$, we can write the mean and variance of a random variable as

$$\mu = \mu'_1 \quad \text{and} \quad \sigma^2 = \mu'_2 - \mu^2.$$

We can generate all moments from a moment-generating function.

Example 3: Create a function in *R* to calculate *r*th moment for probability density function $f(x) = \lambda e^{-\lambda x}$, $x > 0$ and $f(x) = 0$ elsewhere. Hence find the mean and variance of the random variable X .
Sol.

```
pdf=function(x, lam)
{
pdf=lam*exp(-lam*x)
pdf
}
expec=function(lam, r)
{  int=function(x)
    {
      int=(x^r)*pdf(x, lam)
      int
    }
  expec=integrate(int, 0, Inf)$value
  expec
}
```

After running the file, we can calculate the mean, variance, and third moment as follows:

```
# Mean for lam=2
expec(2,1)
[1] 0.5
# variance for same lam=2 from E(X^2)-Mean^2
expec(2,2)-expec(2,1)^2
[1] 0.25
# Third moment for lam=2
expec(2,3)
[1] 0.75
```

Problems:

- (a) The probability distribution of the binomial random variable X , the number of successes in n independent trials, is

$$f(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n.$$

Find the mean and variance of the distribution by writing a FUNCTION in *R* to evaluate *r*th moment. Take $n = 30$ and $p = 0.4$.

- (b) We say that a random variable has a normal distribution with parameters μ and σ^2 if its density function f is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad x \in \mathbb{R}.$$

Find the first three moments of the distribution by writing a FUNCTION in *R* to evaluate *r*th moment. Take $\mu = 12$ and $\sigma = 4$.